

Are You Talking to Me?

Confidence entanglement in AI-assisted operations

An AI assistant now sits in the path of incident diagnosis, in the seat a second reviewer once held. That arrangement produces a failure worth classifying, confidence entanglement. The assistant occupies the procedural position of an independent check while holding none of the independence, so what it returns reaches the responder as confirmation rather than as review. It rides on confirmation bias, and the arrangement produces it. The defenses a team already has were built for a weak reviewer, somebody who can be argued with, and they are not aimed at a response assembled out of the responder's own account.

The mechanism is a matter of register. When an assistant answers in the voice of a reviewer, the responder tends to grant it a reviewer's authority. An organized answer in that voice can suppress the moment the responder would otherwise have stopped to look for alternatives, because it reads as though the alternatives were already weighed. The working reflex shifts from testing a theory to executing a diagnosis, and nobody has requested an explanation the responder did not raise.

The effect is not sycophancy, and not confirmation bias renamed. A team already knows a room tends to close around the account it walked in with, and it built the second reviewer to counter exactly that. Confidence entanglement differs because what now reflects the framing back sits in the position the second reviewer was there to hold.

Separation, not independence. Independence means somebody who never consumed the responder's framing and is free to introduce evidence the responder did not select. During a live incident that position is rarely available. A team can still build separation, which is a procedure rather than a person, and should not call it independence. Four conditions: Ask for rival causes adversarially. Write down the signal that would tell two of them apart, before opening the dashboard. Check that the signal could come out either way. Do not adopt a cause on evidence sought out because it already fit. Separation does not become independence by being run well.

The assistant is easy to mistake for independence because it holds information the responder does not, in its training and its retrieval. Holding information is not the same as occupying a position from which to challenge. Delivered inside the responder's framing, in the register of a colleague agreeing, that information arrives as confirmation.

The problem sharpens in home-built tooling, where one team designs the prompts, the instrumentation, and the review. Nothing in that loop holds a separate position unless the team put one there, so separation is the most such a team can count on, and only if it builds it deliberately.

A worked example. Consider a latency spike. A responder who has seen something similar enters the reading into the assistant, that it resembles a prior cache-eviction problem. The assistant returns an organized account of why eviction fits. The team ships the cache change, it helps slightly, the incident continues, and the actual cause, a downstream connection pool starving under a retry storm, is never raised. The assistant could have raised it. Nothing in the loop asked it to look past the framing it was handed.

Run the same incident with one step inserted before the fix. The responder asks for the strongest rival causes and the signal that would separate each from the current reading, and records that before opening a dashboard. The result is not clean. One rival is a bad deploy, and there was no deploy in the window, so it is useless. For another, the separating signal is a metric the team does not emit and cannot read. One remains checkable, pool saturation, named before anyone looked. Telemetry then shows the pool at its limit and eviction flat. This does not prove the pool was the cause. Flat eviction counts against the cache reading, though only if that metric covers the affected cache across the window of onset, which is worth confirming before anyone relies on it. Either way a second candidate is on the table, and the closed loop never produced one.

From advice to workflow. The pre-commitment above is workflow design rather than individual skill. Under load, composing a careful prompt and keeping a hypothesis record is the first thing to go while a responder is coordinating others and weighing a rollback. Optional steps are the first to be dropped. The commitment cannot live in a wiki as recommended practice. It belongs in the incident channel as a form, pre-populated by the bot, required before a defined class of hard-to-reverse actions, with a named owner and the evidence at decision time captured automatically. One line is most meaningful: The reading goes down before the responder opens any telemetry. The remainder can be

generated. A break-glass path prevents the form from ever blocking action when delay is the greater danger. The incident commander may proceed at once, with reason and time recorded and a retrospective required.

At the point of action, most decisions select the safest reversible step rather than prove a cause. A cache change that partly helps can be the correct call even when the causal account is wrong. The form therefore asks which action is safe across the rival causes and what would trigger a rollback, not only which cause wins. Separation keeps a team from limiting harm in the wrong direction because nobody questioned the reading.

An unframed second look barely exists once an assistant has read the channel, the ticket, and the runbooks, and once the colleague who is paged has read them as well. What remains is a question of how much separation, not whether. Hand the separate responder the raw signals and withhold the diagnosis, so their read forms before they see anyone else's. That is as much separation as a live incident tends to permit. Use it before a rollback, a data change, or a severity downgrade.

What a team can do. Take one class of irreversible action, place the contract in front of it in the incident channel, populate it by bot, and enforce a single line, that the reading is written before anyone opens the evidence chosen to test it. Add the break-glass path so the gate never becomes the obstacle. On the outputs that turn into decisions, ask whether separation is something the team considers, and whether one look before an irreversible action saw the signals but not the diagnosis.

What this brief does not have. No study sits behind any of it. There is no measurement of how often a responder takes an organized answer for a review, no test of whether register is the cause, and no evidence that the form improves how an incident comes out. The argument is from structure, and an argument from structure can show where a failure has room to happen without showing how often it happens.

Two edges stay rough. The class of hard-to-reverse actions is drawn by hand, by a team that knows its own system, and nothing here tells you where to draw it. The account also treats the failure as a single moment, which is not how an incident runs; a reading that sets in early is the one the team is still carrying an hour later, when questioning it costs the most.

None of that changes what a team can do on Monday. The form costs review time, and it can fail in the same way everything else here fails, by being filled in and counted as a review that nobody performed. Watch for that in the first month

of running it.

Measuring it is the next piece of work, and I am setting that up now. It needs incident responders willing to spend twenty minutes on a vignette, and a team willing to run the gate in one channel for a quarter. Even an off-the-record conversation about how your channel actually runs is useful.

brian.moriarty@stevens.edu